

AIM: Build a simple Merkle Tree using hash function

```
const crypto = require("crypto");
function sha256(data) {
  return crypto.createHash("sha256").update(data).digest("hex");
}
function buildMerkleTree(dataArray) {
  if (!dataArray || dataArray.length === 0) {
    return null;
  }
  // Create leaf hashes
  let currentLevel = dataArray.map(item => sha256(item));
  console.log("Leaf Hashes:", currentLevel);

  // Build the tree until one hash (Merkle Root) remains
  while (currentLevel.length > 1) {
    let nextLevel = [];

    for (let i = 0; i < currentLevel.length; i += 2) {
      const left = currentLevel[i];
      const right = (i + 1 < currentLevel.length)
        ? currentLevel[i + 1]
        : left; // Duplicate last hash if odd number of nodes

      const parentHash = sha256(left + right);
      nextLevel.push(parentHash);
    }

    currentLevel = nextLevel;
    console.log("Next Level Hashes:", currentLevel);
  }

  return currentLevel[0];
}

const transactions = ["Tx1", "Tx2", "Tx3", "Tx4", "Tx5"];
//const transactions = ["HELLO WORLD"];
//const transactions = ["HELLO", "WORLD"];

console.log("Original Data:", transactions);

const merkleRoot = buildMerkleTree(transactions);
console.log("Final Merkle Root:", merkleRoot);
```

```
C:\Users\Admin\Desktop\tycs34>node Merkeltree.js
Original Data: [ 'Tx1', 'Tx2', 'Tx3', 'Tx4', 'Tx5' ]
Leaf Hashes: [ '55f743d0d1b9bd86bbd96a46ba4272ddde19f09e3f6e47832e34bb2779a120b5',
  '80ed43f7a11b3295850dd90cc0cfc9a80334f433af8d3d88a1c5e78aff14988f',
  '13288c2ba4bbc9af05aa9ccd39b0cc603dc9e30471d97565c9ef3c3604b7ca23',
  '75af2038a4bcf0230372d08b917047bdcbad80e5f130061bd5b31596df174b67',
  '68ef264fc4289f52448d9d44991745c9be7ef3f0216bf66a580f4edb69543bdd' ]
Next Level Hashes: [ '0971909734e9c49e0f45caeb15a450d717de387a0a27df245e7e924bb7e62b0e',
  '2b4b1ceff3308595e26df24646583f4e3b38f4448c692ab80558f69fea74a3d7',
  '7bd3c22e411418b1d5b255212ffe136f9d04651550f2cc457210a4bd5fa3e7803' ]
Next Level Hashes: [ '5b260dbcbff182d10cdbd21d8cb9e4446fe71820bb91c8dc8dcd0e8a9c8ac',
  'effd0284e2f903f6c1d11236259e5619b448dd706d6dd2729b63e077c1df3585' ]
Next Level Hashes: [ '92c270ef928c4b5c138fa4122c226dcd75ca98cfe1230497c90a8016f4952b11' ]
Final Merkle Root: 92c270ef928c4b5c138fa4122c226dcd75ca98cfe1230497c90a8016f4952b11
```

const transactions = ["HELLO WORLD"];

```
C:\Users\Admin\Desktop\tycs34>node Merkeltree.js
Original Data: [ 'HELLO WORLD' ]
Leaf Hashes: [ '787ec76dcafd20c1908eb0936a12f91edd105ab5cd7ecc2b1ae2032648345dfff' ]
Final Merkle Root: 787ec76dcafd20c1908eb0936a12f91edd105ab5cd7ecc2b1ae2032648345dfff
```

const transactions = ["HELLO","WORLD"];

```
C:\Users\Admin\Desktop\tycs34>node Merkeltree.js
Original Data: [ 'HELLO', 'WORLD' ]
Leaf Hashes: [ '3733cd977ff8eb18b987357e22ced99f46097f31ecb239e878ae63760e83e4d5',
  'd7b0bbea3a935222c4198c38e30b2eb3e111d11dea87fa53547eac1c8a4ff03b' ]
Next Level Hashes: [ '743326a599ecf06117e28339440f6a6d52bf3f2b512ae77e9d8d4adac36a5c97' ]
Final Merkle Root: 743326a599ecf06117e28339440f6a6d52bf3f2b512ae77e9d8d4adac36a5c97
```